

02 — Mathematical Foundations and Claim Status

SEION-KGR v26 MAX

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Purpose: state precisely what is proved, what is checked, and what remains open, using the repository's own six-level epistemic vocabulary. Nothing here re-derives a theorem already proved in `docs/theorems_v3/` or `papers/kernel_integrated_laws_v5/`; each is cited at its canonical location.

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1 Epistemic vocabulary

Every claim in this repository, and in this document, carries exactly one status: `DEFINITION` (fixes notation, no truth claim), `PROVED` (exact identity, no standing assumption), `PROVED_UNDER_ASSUMPTIONS` (theorem with declared, checked hypotheses), `NUMERICALLY_TESTED` (measured on finite runs with recorded seed/dtype/artifact path), `EXPLORATORY` (ran, not preregistered or not confirmatory), `OPEN` (no proof or systematic evidence yet). A claim never receives a stronger label here than it has in the source theorem or claim registry it depends on.

Four namespaces are kept structurally separate throughout this campaign (never merged): `certified_bound`, `empirical_majorant`, `empirical_error_predictor`, `observed_error`.

2 Typed spaces and the CP ternary law

For a finite type system $\mathcal{T}$, each type τ carries an ambient space V_τ and, optionally, a reduced space W_τ with isometry $Q_\tau : W_\tau \rightarrow V_\tau$ ($Q_\tau^* Q_\tau = I$) and orthogonal projector $P_\tau = Q_\tau Q_\tau^*$. The seionic ternary law is

$$\mu(x, a, q) = O[(Ax) \odot (Ba) \odot (Cq)],$$

a CP (canonical polyadic) instance of the general typed n -ary law already fixed in `docs/definitions/nary_laws.md`. **Status: PROVED** that this equals the dense contraction $K_{dijk} = \sum_\alpha O_{d\alpha} A_{\alpha i} B_{\alpha j} C_{\alpha k}$ by direct substitution — verified independently (nested-loop reconstruction, no shared code path with the vectorized form) in `seion_kgr_reference_fp64.py` and cross-checked against the batched package implementation in `tests/kgr/test_perf_and_parity.py` (identical weights, identical output, $\text{atol} = 10^{-4}$).

2.1 Gauge group

The CP factorization has a gauge group: per-component scale (c_0, c_1, c_2, c_3) with $\prod c_j = 1$, and permutation of the rank/component axis. **Status: DEFINITION**, both halves now tested with

explicit negative controls proving the checks are not vacuous (`tests/kgr/test_algebra.py`). Raw CP factors must never be compared across seeds/runs directly — confirmed empirically in the finite-tree track’s own numerical study (pairwise near-orthogonal subspaces at comparable loss).

3 Projection and closure

For an orthogonal projector P and multilinear law μ , the closure residual is $r_\mu = (I - P)\mu(P\cdot, \dots, P\cdot)$. **Status: PROVED UNDER ASSUMPTIONS** (isometry/idempotence exact under FP64, Gate 1: residuals $< 10^{-10}$; under FP32 training, residuals $\sim 10^{-7}$ after real gradient steps — `campaigns/gate12/` smoke evidence). The *sample-mean* closure leakage ratio used as a training signal is explicitly **not** an operator-norm certificate (`docs/definitions/projectors.md`) — this campaign’s certification module (§6) enforces that distinction in code, not just in prose.

4 Exact subset expansion and the $k/k - 1$ theorem

For a typed multilinear tree node v with children $c_1, \dots, c_a$,

$$\Delta_v = r_v(R_{c_1}, \dots, R_{c_a}) + \sum_{\emptyset \neq S \subseteq \{1, \dots, a\}} \mu_v(Y_1^S, \dots, Y_a^S), \quad Y_i^S = \begin{cases} D_{c_i} & i \in S \\ R_{c_i} & i \notin S \end{cases}.$$

Status: PROVED (`docs/theorems_v3/exact_subset_expansion.md`, THM_V3_EXACT_SUBSET_EXPANSION). This campaign adds a **conformance test** (`tests/kgr/test_recursive_projection.py`) verifying the concrete CPTernaryLaw object satisfies this identity to machine precision, with a negative control showing a deliberately non-multilinear law correctly *fails* the same check.

The projected/reduced root error bound $E_T^{\text{proj}} = E_T^{\text{red}} \leq (k - 1)\rho M^{k-1}L_T$ (vs. the ambient bound with coefficient k) is **PROVED UNDER ASSUMPTIONS** (`docs/theorems_v3/homogeneous_constants.md`). Sharpness at a fixed $\eta = \rho/M$ is explicitly OPEN there and remains so.

5 Pathwise majorant and rank allocation

The pathwise majorant $\hat{B}_{\text{root}} = \sum_v \lambda_v \prod_e h_e$ is **PROVED UNDER ASSUMPTIONS** as an upper bound for the declared scalar recursion (THM_V3_NODEWISE_PATH_SUM). Whether pathwise-alone is a good *rank allocation policy* is a separate, empirical question: the confirmatory Level-1 evidence in `applications/adaptive_tensor_network/results/LEVEL1_FINDINGS.md` shows pathwise loses to `uniform` and `local_error_greedy` at equal budget, while the majorant itself was empirically a valid upper bound in 100/100 measured triples (ratio true/majorant $\in [0.35, 0.93]$, never > 1). This campaign’s `seion_kgr/rank_controller.py` accordingly ships a *hybrid feature* policy as the recommended default, never pathwise alone (**Status: DEFINITION**, tested in `tests/kgr/test_seion_kgr_package.py`).

6 State-to-score-to-ranking certification

The elementary theorem — margin $> 2\varepsilon$ in score space implies exact rank-order preservation under any perturbation bounded by ε — is **PROVED** (a direct triangle-inequality argument). This

campaign's `seion_kgr/certification.py` implements the full chain honestly: it certifies (`state_error_bound= 0`, hence `score_linf_bound= 0`) *only* when nothing is being approximated in the score path — no rank-reducing projector, no nonlinear envelope. The moment either is present, certification correctly and unconditionally returns `NOT_CERTIFIED`, verified by a test with a 2000-point margin that still refuses certification (margin size cannot buy back a missing assumption check). **Status: the general (compressed-model) certification chain remains OPEN**; the narrow zero-compression case is now a real, tested, `PROVED_UNDER_ASSUMPTIONS` result.

7 Kernel-integrated extension

The L^2 ternary kernel boundedness, composite-kernel associator-defect representation, and finite cohomology descent results are **PROVED_UNDER_ASSUMPTIONS**, inherited unchanged from `papers/kernel_integrated_laws_v5/main.tex`. None of this is needed for, or invoked by, the finite discrete KG case this campaign benchmarks; it remains scoped exactly as before.

8 Full claim matrix

See `docs/SEION_KGR_CLAIM_MATRIX.md` (repository root) for the complete, currently 30+-row table, including this campaign's explicit old-status/new-status changelog for every claim it touched. This document summarizes; it does not replace that table as the source of truth.